Junjie **Yu**

AI Researcher | NeuroScience / Learning Theory / Foundation Models

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Né le 23 septembre 1982 (35 ans) à Nouméa, Nouvelle-Calédonie



Ph.D. student working at the intersection of foundation models, representation learning, and learning theory. My research focuses on understanding how the geometry and structure of learned representations shape generalization, transfer, and alignment across tasks and domains. I have developed and analyzed foundation models for EEG and studied representation geometry in large-scale neural networks, including the connections between pretrained representations, downstream adaptation, and generalization. My work also investigates the convergence and alignment of representations across artificial neural networks and the human brain using EEG and fMRI. I am particularly interested in foundation models, multimodal learning, representation learning, and the theoretical principles underlying scalable and transferable AI systems.

Education

2022–Present	Ph.D. Student (Advisors : Prof. Yi Guo and Prof. Quanying Liu), Southern University of Science and Technology, China
2019–2021	Master of Applied Actuarial Studies, Macquarie University, Australia
2014–2018	B.Sc. in Mathematics and Applied Mathematics, South China Agricultural University, China

Experience

2021–2022	Visiting Student & Research Assistant (Advisors : Prof. Quanying Liu), Neural Computation and Control Laboratory, Southern University of Science and Technology, China
2019–2020	Algorithm Intern, Shenzhen Academy of Urban Public Safety and Technology, China
2018	Data Analyst, Dongguan Tong, China

Selected Projects

Representation Geometry and Generalization Theory — 2026

https://icml.cc/virtual/2026/poster/62049

- Developed a representation-based generalization framework linking embedding dimensionality, Wasserstein distribution convergence, and generalization error under finite-sample learning.
- Proved a generalization error upper bound showing that lower-dimensional representations reduce the discrepancy between empirical and population distributions, improving generalization when the model sufficiently fits the training data.
- Extended the framework to deep neural networks and demonstrated that the dimensionality of the final-layer embedding provides a practical predictor of generalization performance, while accounting for the effect of network Lipschitz continuity.

Representation Learning | Wasserstein Distance | Generalization Theory

Neural Representation Geometry and Learning Performance — 2026

https://www.sciopen.com/article/10.26599/TST.2026.9010025

- Analyzed whole-brain fMRI representations collected during online learning to investigate the relationship between neural representation dimensionality, Wasserstein distance, and individual learning performance.
- Revealed a significant relationship between representation dimensionality and Wasserstein distance across widespread cortical regions, consistent with the proposed representation-based generalization framework.
- Found that lower-dimensional representations in higher-order association regions, including the angular and supramarginal gyri, were associated with better learning performance, while visual, memory, and semantic processing regions exhibited distinct positive relationships.

fMRI | Representation Learning | Statistical Analysis | Wasserstein Distance

Convergent Representation Geometry in AI and the Human Brain — 2026

https://icml.cc/virtual/2026/poster/65699

- Investigated whether high-performing neural networks and the human brain converge toward similar representational geometries using shared visual stimuli and fMRI responses across multiple vision models.
- Established systematic relationships among AI–AI alignment, AI–brain alignment, and model generalization performance, showing that models with better brain alignment also tend to exhibit stronger inter-model alignment and generalization.
- Demonstrated that embedding dimensionality, particularly local intrinsic dimensionality, provides a unified geometric predictor of AI–AI alignment, AI–brain alignment, and generalization performance.

LLM | fMRI | Representation Alignment | Intrinsic Dimension | Generalization

Stable Parameter Subspaces for Transferable Neural Networks 2025 - 2026

https://arxiv.org/abs/2605.07302

- Investigated the geometry of parameter updates during downstream fine-tuning across diverse vision, language, EEG, and fMRI foundation models.
- Discovered that fine-tuning consistently preserves a stable low-dimensional parameter subspace established during pretraining, even under substantial parameter updates and across different downstream tasks and distributions.
- Showed that larger-scale pretraining produces more stable and transferable parameter subspaces, and demonstrated that restricting adaptation to these subspaces enables effective parameter-efficient fine-tuning with only 0.2% trainable parameters.

Python | Deep Learning | Foundation Models | Parameter-Efficient Fine-Tuning | Representation Geometry | Spectral Analysis

Understanding and Correcting Low-Frequency Bias in EEG Foundation Models 2025 - 2026

https://arxiv.org/html/2608.01898v1

- Investigated spectral bias in EEG foundation models and discovered a persistent preference for low-frequency information across diverse pretraining objectives, dataset scales, and model capacities.
- Revealed that the interaction between EEG's $1/f^{\alpha}$ spectral structure and neural networks' low-frequency learning preference, together with power-weighted reconstruction losses, systematically limits the representation of high-frequency neural dynamics.
- Proposed FAME, a frequency-balanced masked autoencoding framework that independently standardizes time–frequency targets and equally weights frequency-band reconstruction losses, promoting more spectrally balanced and transferable EEG representations.
- Demonstrated strong cross-task generalization on 41 downstream EEG tasks, with the 1B-parameter FAME model achieving state-of-the-art performance on 24 tasks and showing that reduced frequency bias is associated with improved downstream transfer.

Python | Deep Learning | Foundation Models | Parameter-Efficient Fine-Tuning | Representation Geometry | Spectral Analysis

Publications

2026 **Yu, J.**, Ma, W., Wei, C., Zhang, J., Deng, H., Deng, Z., & Liu, Q. Local Intrinsic Dimension of Representations Predicts Alignment and Generalization in AI Models and Human Brain. *Proceedings of the International Conference on Machine Learning*.
CONFERENCE | CCF-A | ICML 2026 ICML

2026 **Yu, J.**, Ouyang, Z., Deng, H., Wei, C., Ma, W., Zhang, J., …& Liu, Q. Understanding Generalization from Embedding Dimension and Distributional Convergence. *Proceedings of the International Conference on Machine Learning*.
CONFERENCE | CCF-A | ICML 2026 ICML

2026 **Yu, J.**, Ma, W., Pei, Z., Zhu, L., Shi, X., Liu, Y., …& Guo, Y. Evaluating the biological meaning of neural network decisions in EEG-based MCI detection. *Journal of Neural Engineering*, 23(4), 046003.
JOURNAL | SCI | JCR Q2 DOI

2026 **Yu, J.**, Deng, Z., Wei, C., Ma, W., Zhang, J., Deng, H., …& Liu, Q. Low-Dimensional Representations Support Efficient Learning Across Brains and AI. *Tsinghua Science and Technology*.
JOURNAL | SCI | JCR Q1 DOI

2022 **Yu, J.**, Li, C., Lou, K., Wei, C., & Liu, Q. Embedding decomposition for artifacts removal in EEG signals. *Journal of Neural Engineering*, 19(2), 026052.
JOURNAL | SCI | JCR Q2 DOI

Preprints

2026 **Yu, J.**, Deng, Z., Zhang, J., Mu, J., An, J., Ma, W., …& Liu, Q. Understanding and Correcting Low-Frequency Bias in EEG Foundation Model.
PREPRINT arXiv

2026 **Yu, J.**, Zhang, J., Pei, Z., Shi, X., Liu, Y., Jiang, X., …& Guo, Y. Complexity and Stability of Neural Activity Across Aging and Neurodegenerative Disease.
PREPRINT arXiv

2026 **Yu, J.**, Wang, Y., Deng, Z., Zhu, Y., Ma, W., & Liu, Q. Pretraining Induces a Reusable Spectral Basis for Downstream Task Adaptation.
PREPRINT arXiv